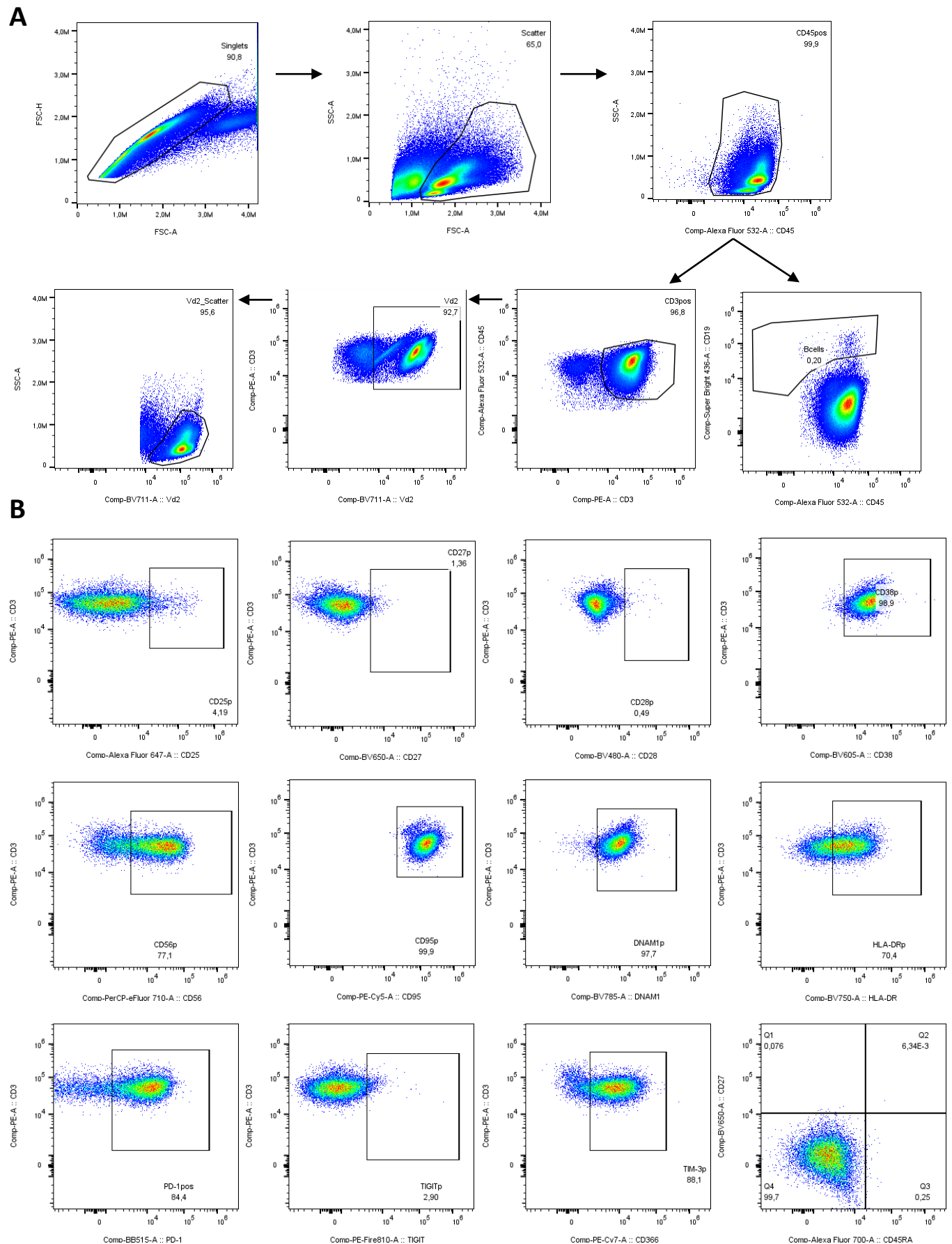
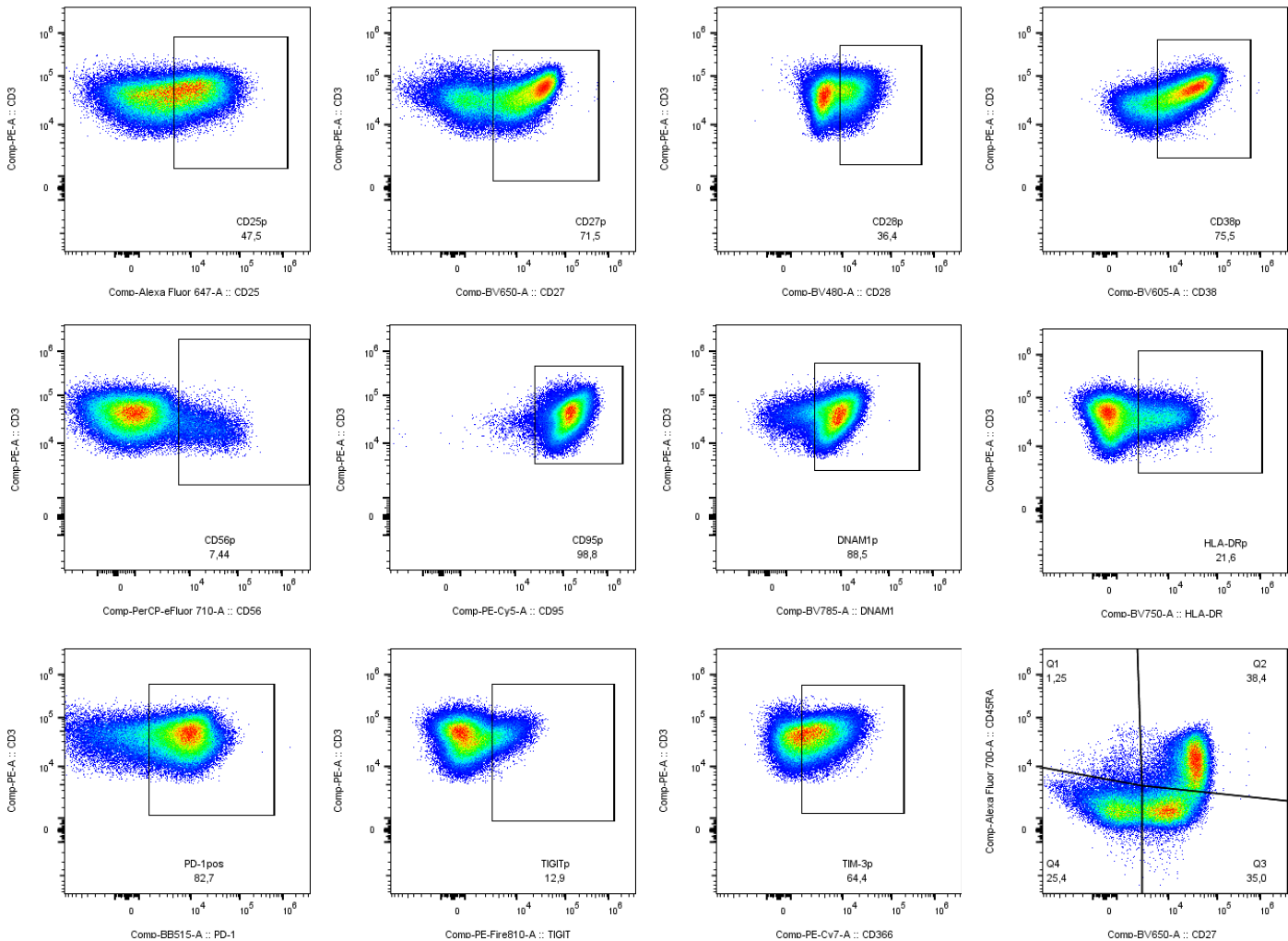
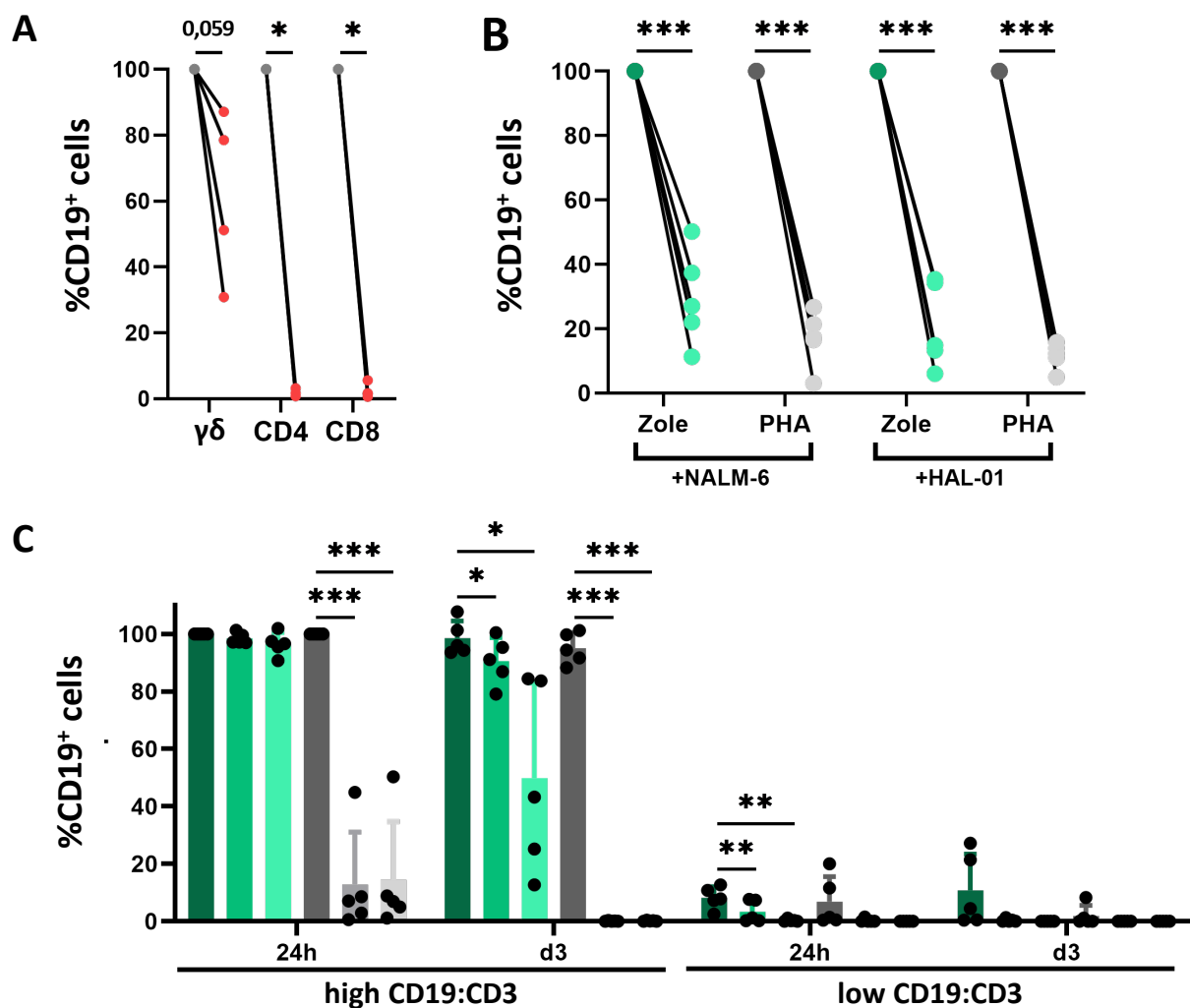




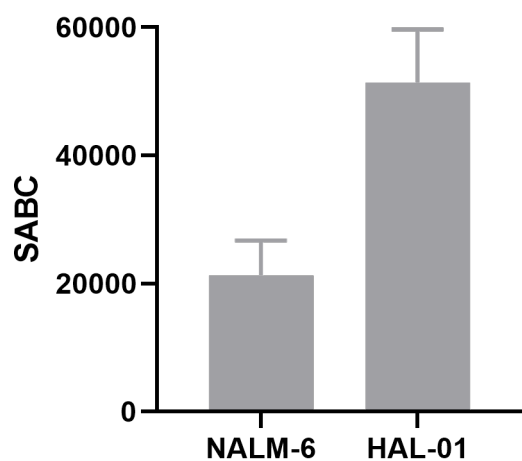
B



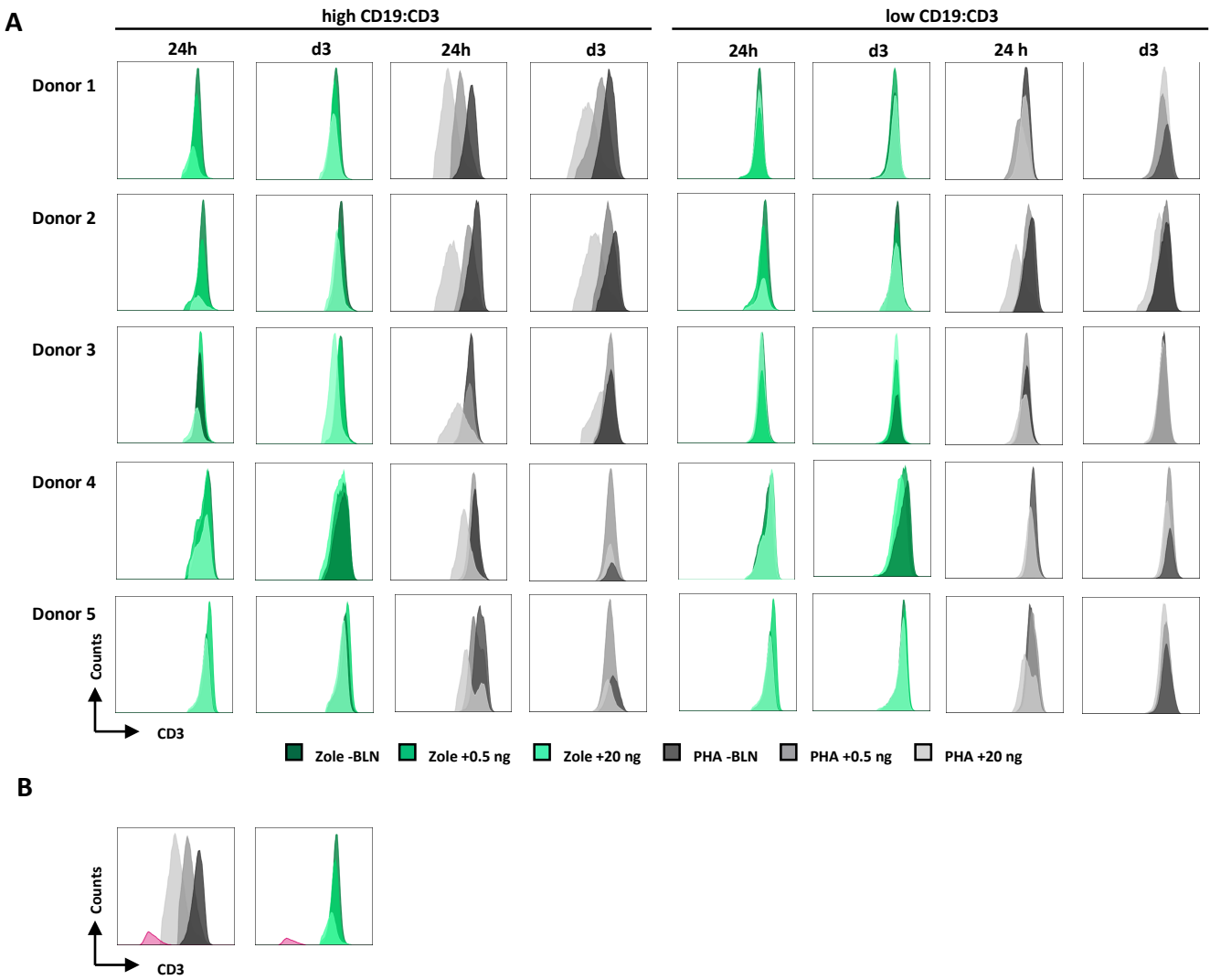
Supplemental Figure 2. Flow cytometry gating strategy for PHA-expanded $\alpha\beta$ T-cell cultures. (A) Representative backbone gating (singlets, viable cells by FSC/SSC, lymphocytes by CD45-expression and T and B-cells. (B) Representative phenotyping strategy for CD8+ $\alpha\beta$ T cells and downstream subset/marker gates used for analyses.



Supplemental Figure 3: BLN-specific lysis of CD19⁺ cells in in vitro cultures. Data correspond to Figure 1A (A), Figure 2A (B), and Figure 3A (C). For each donor, residual CD19⁺ cells in the -BLN condition were set to 100%, and the corresponding +BLN condition was expressed relative to this baseline: normalized residual CD19⁺ (%) = $100 \times (\text{residual CD19+ \% +BLN}) / (\text{residual CD19+ \% -BLN})$. (Equivalently, "BLN-specific lysis" can be derived as $100 - \text{normalized residual CD19+ [\%]}$.) (A) Freshly isolated CD4⁺, CD8⁺, and $\gamma\delta$ T cells co-cultured with HAL-01 for 3 days (n=4 donors). (B) PHA-expanded $\alpha\beta$ and Zole-expanded $\gamma\delta$ T-cell lines co-cultured with HAL-01 or NALM-6 for 24 h at low target burden (E:T 5:1; low CD19:CD3) (n=5 donors). (C) Expanded $\alpha\beta$ and $\gamma\delta$ T-cell lines co-cultured with HAL-01 under low target burden (E:T 5:1) and high target burden (E:T 1:5) and treated with BLN (0.5 or 20 ng/mL) for 24 h and 3 days (n=5 donors). Donor-matched $\pm$ BLN comparisons were analyzed using paired two-tailed t-tests. *p<0.05, **p<0.01, ***p<0.001.

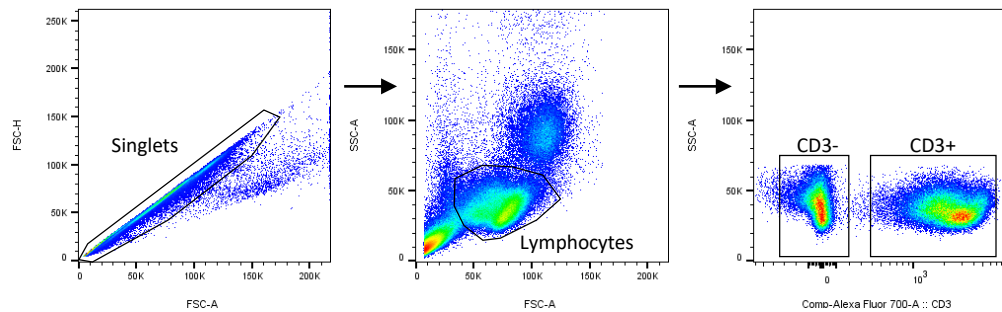


Supplemental Figure 4: Quantitative determination of CD19-antigen expression on B-cell surface of HAL-01 and NALM-6 cell lines as analyzed by flow cytometry using the purified anti-human CD19 antibody (clone HIB19) and QIFIKIT®. Shown are the median specific antibody binding capacity (SABC) values + 95% CI of four experiments.

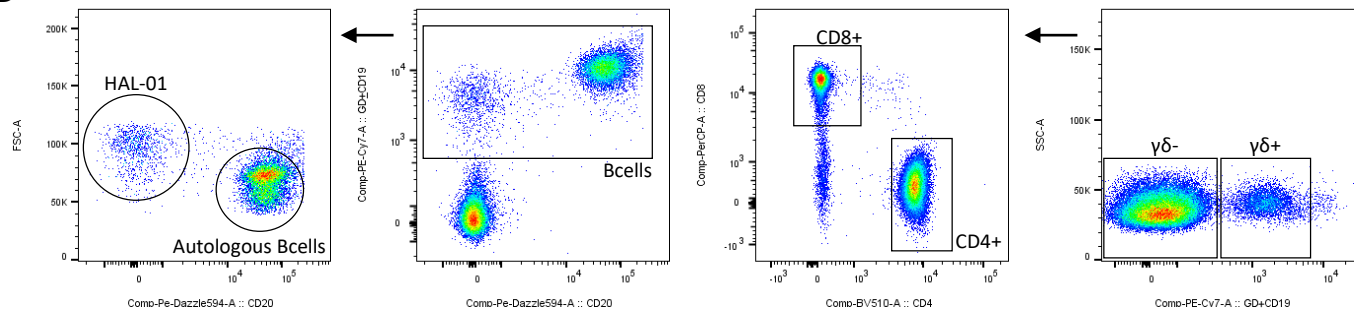


Supplemental Figure 5. CD3 down-modulation depends on CD19 target burden and is prominent in expanded $\alpha\beta$ but not $\gamma\delta$ T cells. (A) CD3 expression (MFI) in PHA-expanded $\alpha\beta$ and Zole-expanded $\gamma\delta$ T-cell cultures treated with BLN (0.5 or 20 ng/mL) under low target burden (E:T 5:1; low CD19:CD3) or high target burden (E:T 1:5; high CD19:CD3) after 24 h and 3 days (n=5 donors). (B) Representative CD3 histograms from PHA- or Zole-expanded cultures shown alongside the matched unstained control (pink).

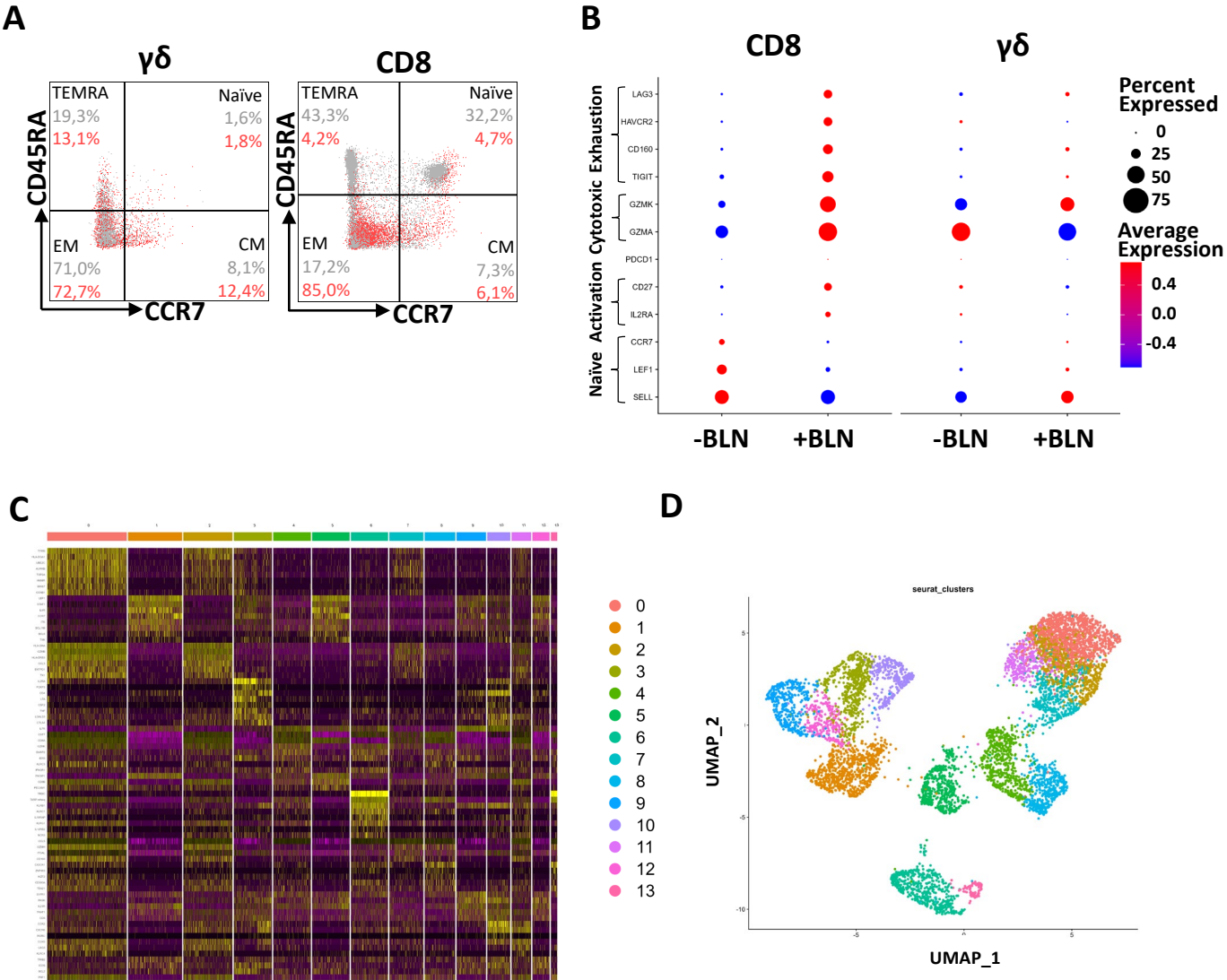
A



B



Supplemental Figure 7. Flow cytometry gating strategy for PBMCs co-cultures. (A) Representative backbone gating used throughout the study (Singlets, lymphocytes by FSC/SSC, live cells), followed by identification of CD3+ T-cell subsets and target populations (HAL-01 cells and CD19+ B cells).



Supplemental Figure 8. Supportive analyses for targeted single-cell RNA-seq of BLN-stimulated PBMC cultures (single donor). PBMCs from one healthy donor were stimulated in vitro $\pm$ BLN prior to CD3⁺ T-cell isolation and targeted scRNA-seq. (A) Flow-cytometric immunophenotyping of $\gamma\delta$ and CD8⁺ T cells in treated vs untreated cultures; naïve, CM, EM, and TEMRA subsets were distinguished by CCR7 and CD45RA. (B) Dot plot of selected T-cell fitness/exhaustion-associated genes in CD8⁺ and $\gamma\delta$ T cells; dot size indicates the percentage of cells expressing the gene and dot color indicates average expression. (C–D) UMAP visualization of Seurat-derived clustering (14 initial clusters) prior to manual curation, annotated as: 0: CD8⁺ proliferating; 1: CD4⁺ naïve; 2: CD8⁺ proliferating; 3: CD4⁺ EM; 4: CD8⁺ TEMRA; 5: CD8⁺ naïve; 6: $\gamma\delta$ ⁺; 7: CD8⁺ EM; 8: CD8⁺ TEMRA; 9: CM; 10: CD4⁺ EM; 11: CD8⁺ EM; 12: CD4⁺ CM; 13: $\gamma\delta$ ⁺